Ok, hello.

This talk is about …. The *big* topics. Number 1 – scaling to 8 billion users. Number 2 – offering privacy to those users. And – Number 3, accomplishing this without a soft-fork/hard-fork – actually, it is even better than this. The truth is it does not require a pull request to Bitcoin Core, or any code changes to Bitcoin Core – which is different, but better. As we all saw, recently, with the OP Return incident – even a lowly pull request is actually a non-starter -- -- even for something that is *not* a soft fork [or hard fork] [or anything]. In contrast, to that, the Ordinals-2023 phenomenon has shown that – even a very very unpopular and contentious thing , is viable, as a separate piece of software , "on top" of the Bitcoin Core stack.

So – that's quite the claim – quite audacious. In fact, at PlebFi 2023, I had this slide – and I presented on how Bip-300 can actually solve this *whole long list* of problems. We don't have time for this giant list. But the claim is very bold -- so why should you take this seriously?

Well first, we actually have quite a few endorsements. More than 50 – . You can check them out, at LayerTwoLabs.com/friends . [[Yes – we are the stage sponsor, this year]] We have an Adam Back one – we have a fiatjaf one – fiatjaf is the creator of nostr , over next door. We have a Robin Linus – creator of BitVM , and ZeroSync (and other stuff). We actually have a few quote from each of these names. – Um, so that's *a start.*

A second reason is that the software is out! We have test , play-money software. ((Of course – we had TEST software in the past – but that required the Bitcoin Core pull request – the new software does not. – Although it is still with fake money – on SigNet.)) Try it yourself. Go to layertwolabs.com slash download. And try it out. – But, don't take my word for it – form your own experience. Here it is! We have some screenshots. So, check it out. [[~]]

A third reason, is that the idea is very *fleshed out* – I have *consistently* been explaining – (in detail) – exactly how it works, and why it works – why it will work – over and over again for many years.

On the scaling side -- -- In November 2015 – I explained how Drivechain would allow us to resolve the Blocksize War – by having both large AND small blocks, simultaneously, (on different networks). -- -- In October 2016 – I presented about this, at Scaling III – in Milan – shrinking the L1 blocksize, and scaling via optional L2s with larger blocks. In March 2018 , February 2021 and again (last year) May 2024 , I gave concrete -- actual examples –numbers -- and parameters – for how we would (eventually) process 9 trillion – trillion transactions per year, and send all that fee revenue to the miners. (automatically)

On the privacy side – back in April 2020, --[long before anyone had heard of Michael Saylor]— we organized a contest – [this was before LayerTwo Labs , and it involved some different people, but I put it together] – this contest for a privacy L2. -- -- In March 2021, we put out *a video* – a demo video, of the *completed software* working – with a User Interface, and buttons you can click, etc.

And of course I have presented about all of this, at this conference [[the Bitcoin Magazine Conference]], many times – for example, four years ago – June 2021, in the big tent – there was that huge tent – you can watch that talk if you want. [[It was a pretty good talk]] -- -- -- So, it has been around for a while.

We also have our *critics* -- (just like anyone else). We reply to each criticism at enormous length. – I think -- One reason to be *optimistic* about this project, is because our critics never produce anything good. For example, Peter Todd was a critic of ours – he wrote this long post, as a criticism – which is great. But then, I could reply to it in detail – and it turns out that almost every single sentence and paragraph of Peter's article, was incorrect. [[I replied to it in this giant image here – which you can check out for yourself, when you have more time]] And actually, -- furthermore --, Peter was on stage at Baltic Honeybadger 2023, at the same time he was writing his critique article – and he said – [on stage] – that our work was just in the idea phase, and there was no code written for this.

The problem with that, of course, is that it's not true – at all. This download link, has been at the top of the main page, drivechain.info , for *years*. The privacy L2 video - -that I had mentioned before, not only code ; but with a graphical user interface [[with buttons that you can click]] – *that* video was two-and-a-half years old – when Peter said that on Baltic Honeybadger stage. So – I just bring that up to say -- we do have critics – but they never read the BIPs (which are only like 5-6 pages by the way) – and they've never *visited our website*, or tried the software – (even as a user). Like they haven't clicked the buttons – let ALONE , examining the code. Or constructing some kind of informed argument. People don't even watch our youtube videos on the main page – which is like the lowest effort ever.

So you have to set that against – set the critics, against all the endorsements (and supporters), we have. I mean – anyone can just *be* a critic. And – especially if an idea is really good – you may be *threatened* by it, and want to criticize it even more.

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Ok, now --- the remainder of the talk, [what I would like to do is], -- Three things: [Number 1] compare our software (and ideas) , with the current state-of-the-art, in Bitcoin. [Number 2] give a technical argument why only the *merged-mined-L2*s will survive – such as ours. In contrast, the non-mined L2s – [such as Lightning Network, and ARK] – will not survive. And [Number 3] I will give a little technical explanation – of how this is all accomplished, without a pull request to Bitcoin Core [in other words, without changing Bitcoin Cores codebase at all].

[9 minutes]

Okay I am about to compare the Drivechain idea of scaling, with the Lightning Network's idea of scaling. First, what is my idea of scaling? Well here I have hypothetical projections of Bitcoin taking over the planet Earth – over 10 years. If you divide the world into 13 regions, then the second-to-last column has the L2-blocksize requirements.. which are 12.9 MB rising to 2.3 GB – so the plan would be to have 13 regionally-distributed largeblock L2s with those blocksizes. [[Again these are optional L2s, not on Layer 1 – so we can set the Blocksize to whatever we want.]] Even with today's hardware I have estimate this cost to be only a few hundreds dollars per money – and it is certainly going to plummet over time, both in nominals terms , and as a percentage of people's incomes.

Anyway, here's the comparison. [[and these tables, are on LayerTwo Labs com , so you don't have to read them now]] . Onboarding – Lightning does not help with onboarding at all. Just for that reason alone – we should give up on the lightning network. Anyway, with the 13-13 strategy I just outlined, you can onboard 8 billion people, directly to the L2. Liquidity – in our idea, there is no liquidity – its always infinite. In LN, you are screwed by liquidity constantly. In Lightning Network, the payments fail, sometimes. In ours there's no failing. In Lightning, the bigger the payment is, the more likely it is to fail, or have a high fee, or whatever. I'm going to talk about the "fee pass-through" idea in the next section.

[[read slides]]

[12:00]

~Motivation/Endgame~

Ok -- this is the endgame for Bitcoin as I see it. [[This giant circle]] SO -- these circles are to scale – the left circle is total transaction fees paid to miners in 2022. The next one over, is fees in 2023 – the ordinals year. The third circle is all transaction fees paid by all Altcoins (2022). [[Ethereum, etc]] And the big circle is the value of all txn fees paid on Earth (2022). Ok so – this my estimation.

Now – how did I get this number? 6.4 trillion txns/year , times $0.10 per txn. – [[I wrote a big article about it]] You can read the details if you want. . It appears to be growing at a rate of 6.2% per year – which would mean it doubles -- the area of this circle doubles every 11-12 years.

The reason I bring this up, is to ask a very important question: if Bitcoin took over the world, in 10 years (as I was mentioning), then ***who would get this money?*** The current intellectual climate, is (incorrectly) pushing non-mined L2s. -- -- which include: The Lightning Network – ARK – Rollups -- Liquid. All of these L2s are not mined L2s – the txn fees do not go to miners, instead they go to someone else -- the lightning node , or the LSP -- or the ARK-SP -- or the sequencer (the rollup sequencer). The money goes to someone else.

Now, first of all -- this is kind of rude to the miners. But it's way worse than that.

~Instability~

It produces a horrendously unstable situation, that will end in: 1) conflict (between L1 and L2), 2) huge incentives for miner-centralization, and then eventually, 3) vertical integration between the L2s and the L1 miners, and this vertical integration will 4) utterly obliterate the entire L2 security model, in the first place – which means that all the code [[that people]] plan to write, about these other L2s, is a waste of time. The L2 will end up fully custodial , possibly even miner-custodial. The value will all become MEV. Technically.

Ok – let me explain slightly. So -- the miners get this small circle [[at the bottom there]] , not the large circle, but what can they do about it?

Unfortunately: everything. Well , every L2 is vulnerable to 51% attack , in some way or another. For example, the Lightning Network requires an L1 txn to join the network, and one to leave the network, and the LN only works at all, if you can broadcast “justice transactions” to L1 whenever you want. And of course, 51% miners could censor these txns. If do not believe me – here is Rene Pickahardt – author of the book Mastering Lightning – he is saying exactly what I am saying, here. It is similar for other L2s – BitVM requires that the miners NOT censor – their version of the justice txn, which basically a transaction proving that someone broke the L2s rules . If L1 miners censor these, then the security model of BitVM no longer applies. And furthermore -- no matter what the L2 operational security model is, the L1 miners can always block the entrance and the exit.

So, the miners choose which L2s survive – and which live. So they’re not going to settle for the tiny circle. Only something that gives them the large circle, is stable. Everything else is like a ticking time bomb.

[[18:00]]

Ok – finally -- doing things without a soft fork.

Well, here is a paper about it! Go and read it if you want. ((What is it about, you ask?))

Basically, the “soft fork” part still happens – but in a second piece of software. That software kicks back “invalidateblock” to Bitcoin Core if the block violates the new softfork rules. Or – it just does anything, to get Core to reject the block or just forget about the block. If 51% hashrate runs that – we call it the <u>*enforcer*</u> . If that happens, then the soft fork activates on the network – even if Bitcoin Core has the exact same source code, exact same software binary, etc.

So you can do all kinds of soft forks, even if Bitcoin Core does nothing – which is what they plan on continuing to do.

Oh also, the software is out – not only for Bip300/301, but also for OP CAT – which is much simpler and easier to understand. [[We should make a CTV one.]]

So here is the proposal – for scaling. Miners run this activator software, we turn on 13 of these L2s – scale achieved.

Same process for privacy – just use the privacy-specific L2.